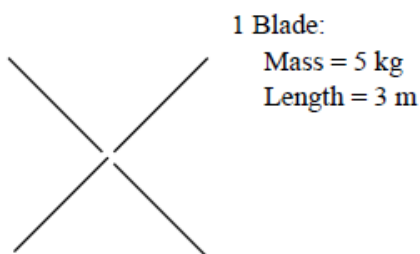


PROBLEMS-Rotational Kinematics

Shown in the figure below is a four blade propeller. Each blade has a mass of 5 kg and a length of 3 meters. The propeller starts from rest and reaches 200 RPM after 20 seconds.

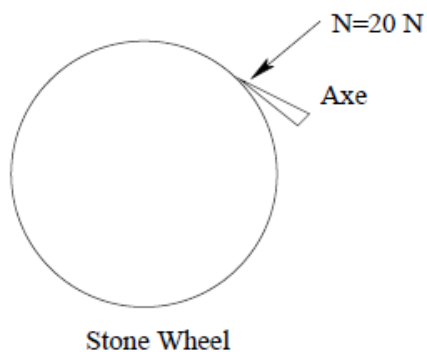


4-Blade Propeller

- (a) What is the angular acceleration of the propeller in $\frac{rad}{sec}$?
- (b) How many turns has the propeller undergone in these 20 seconds?
- (c) The speed of sound in air is $345 \frac{m}{s}$. At what ω does the tip of the propeller exceed the speed of sound?
- (d) Assuming the angular acceleration is constant at what time does the tip of the propeller exceed the speed of sound?
- (e) Approximate the propeller as four rods rotating about their ends. What is the moment of inertia, I , of the propeller?
- (f) What torque is required to make the propeller move as detailed in this problem?

PROBLEMS-Rotational Kinematics

Shown in the figure below is a grindstone (mass = 80 kg, radius = 0.7 m) that is used to sharpen an axe. The axe is pressed against the spinning grindstone with a normal force of 20 N. When this is done the grindstone slows from an initial motion of 100 rpm and comes to a stop in 2 minutes.



- (a) What is the angular acceleration of the grindstone?
- (b) How many turns does the grindstone make before coming to rest?
- (c) What is the moment of inertia of the grindstone?
- (d) What is the coefficient of kinetic friction between the axe and the grindstone?